On the Guidance of Flow Matching: review on synthetic data

Lucas Scherberger
Technical University of Munich
lucas.scherberger@tum.de

Lea Zimmermann
Helmholtz Munich
lea.zimmermann@helmholtz-munich.de

Abstract

Flow matching has shown state-of-the-art performance in various generative tasks in image generation and scientific applicaiton. In many applicatons, it is desirable to generate specific parts of the target distribution which can be dynmaically controlled after training. Guidance methods have been proposed to steer a frozen flow matching model towards a defined objective. In this paper, we evaluate the guidance methods for flow matching on synthetic data and discuss the limiting assumptions of the methods.

1 Introduction

Deep generative models have achieved remarkable success across a wide range of domains, from high-fidelity image synthesis (Rombach et al., 2022) to computational biology (Klein et al., 2025) and drug discovery (Jumper et al., 2021). A shared principle behind these successes is the idea of learning a mapping from a simple, tractable distribution (e.g., standard Gaussian) to the complex target distribution of interest, enabling the model to generate realistic samples by transforming random noise. Diffusion models (Ho et al., 2020; Song et al., 2021; Rombach et al., 2022) realize this by learning to reverse a gradual noise-corruption process, and have set strong baselines across many benchmarks in computer vision. Flow matching (Lipman et al., 2023) approaches the same goal more directly. A time-dependent vector field is defined to continuously deform the prior distribution into the target data distribution, and a neural network is trained to approximate this field from data. At inference, new samples are generated by constructing a trajectory from a point in the source distribution to the target distribution through the learned vector field. This can be achieved by numerically integrating an ordinary differential equation (ODE). A critical design choice is the coupling, that is, how individual source samples from the prior are paired with target samples from the data distribution when constructing the training trajectories. Randomly coupling source and target points produces crossing trajectories that place conflicting demands on the vector field and make learning harder, whereas pairing via optimal transport yields shorter, straighter paths that are easier to approximate and require fewer ODE integration steps at inference (Tong et al., 2024). Entropic variants of this coupling have further extended flow matching to unpaired data settings such as single-cell genomics (Klein et al., 2024), broadening its practical applicability.

In many applications it is desirable to generate not just any sample from the target distribution, but samples that satisfy a user-specified criterion, for instance, images of a particular class. One natural approach is to incorporate this criterion at training time, for example by conditioning the model on a class label or other side information (Ho and Salimans, 2022); the resulting model can then generate samples from the desired subdistribution on demand. However, such training-time conditioning must be built into the model from the start, and obtaining samples under a new or different objective requires retraining.

A more flexible alternative is inference-time steering, commonly referred to as guidance. There, the generation process of an already trained model is modified at inference time to bias sampling toward

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the desired region of the target distribution, without any retraining (Dhariwal and Nichol, 2021; Feng et al., 2025). For diffusion models, this problem is well understood. Classifier guidance (Dhariwal and Nichol, 2021) and classifier-free guidance (Ho and Salimans, 2022) modify the score function at inference time to bias generation toward a desired condition. Transferring these ideas to flow matching is, however, non-trivial, because the deterministic trajectories and the absence of a stochastic noise schedule fundamentally change how an energy function can be incorporated into the velocity field. As a consequence, guidance for flow matching has remained largely underexplored compared to its diffusion counterpart. Feng et al. (2025) recently proposed the first general framework for guidance in flow matching, deriving a family of training-free and training-based techniques covering approximate and asymptotically exact methods. However, a systematic empirical comparison of these approaches and a characterization of their failure modes has not been carried out.

We summarize our contributions as follows:

- We provide a review of the guidance methods for flow matching.
- We experimentally test the guidance methods on synthetic data, visualize their behavior, and compare their performance in terms of different metrics.
- We experimentally examine the limitations of the guidance methods regarding different coupling strengths and desired subdistributions and their effect on the generated samples.

2 Background

2.1 Conditional Flow Matching (CFM)

Flow Matching (FM) is a generative method that learns a time-dependent velocity field $v_t(x)$ (with $t \in [0,1]$ and $x \in \mathbb{R}^d$) that transports a tractable source distribution p_0 (e.g. a standard Gaussian) into a complex target distribution p_1 on $\mathbb{R}^d$ (Lipman et al., 2023). The velocity field $v_t(x)$ is a map with $(t,x) \in [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ since the source and target distributions live in the same space. The velocity field defines a trajectory x_t starting from $x_0 \sim p_0$ through the ordinary differential equation (ODE)

$$\frac{d}{dt}x_t = v_t(x_t), \qquad x_t\big|_{t=0} = x_0. \tag{1}$$

Following this ODE over time from $t=0$ to $t=1$ continuously deforms the source into the target distribution. The distribution of x_t (denoted p_t) coincides with p_0 at $t=0$ and with p_1 at $t=1$. Thus $x_t\big|_{t=1} = x_1 \sim p_1$. A new sample is generated at inference time by drawing $x_0 \sim p_0$ and numerically integrating Eq. (1) forward to $t=1$.

Ideally v_t would be trained by regressing it onto the true marginal vector field $u_t(x)$ that generates the path p_t. This target has no closed form in general. Lipman et al. (2023) sidestep this by conditioning on a latent variable z such as an endpoint pair $z=(x_0,x_1)$ with $x_0 \sim p_0$ and $x_1 \sim p_1$. A conditional probability path $p_t(x|z)$ and a conditional velocity field $u_t(x|z)$ are then chosen in closed form. The marginal path is recovered as

$$p_t(x) = \int p_t(x|z)\, p(z)\, dz \tag{2}$$

and the marginal vector field is the conditional expectation of $u_t(x|z)$ over the posterior $p(z|x_t = x)$ which we can simplify with Bayes' rule to

$$u_t(x) = \int u_t(x|z)\, \frac{p_t(x|z)\, p(z)}{p_t(x)}\, dz. \tag{3}$$

Training v_t to match this marginal target directly amounts to minimizing

$$\mathcal{L}_{FM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, x_t\sim p_t}\left[\|v_t(x_t) - u_t(x_t)\|^2\right]. \tag{4}$$

This expectation is intractable in practice. Evaluating $u_t(x_t)$ requires the integral in Eq. (3). Sampling $x_t \sim p_t$ also requires already knowing the path we are trying to learn. They show that regressing v_t onto the closed-form conditional target $u_t(x|z)$ yields the same gradient with respect to v_t as regressing onto the intractable marginal target in Eq. (4). The marginal objective can therefore be replaced by a tractable conditional counterpart. This yields the simulation-free Conditional Flow Matching (CFM) loss

$$\mathcal{L}_{CFM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, z\sim p(z),\, x_t\sim p_t(\cdot|z)}\left[\|v_t(x_t) - u_t(x_t|z)\|^2\right], \tag{5}$$

where $u_t(x_t|z)$ is the conditional velocity field that transports x_0 to x_1 along the path defined by $z = (x_0, x_1)$. Furthermore, $v_t(x_t)$ is the neural network being trained to imitate the marginal vector field $u_t(x)$. Every variable in Eq. (5) is obtained without ever simulating the ODE in Eq. (1). The time t is drawn uniformly from $[0, 1]$. The pair $z = (x_0, x_1)$ is obtained by sampling $x_0 \sim p_0$ and $x_1 \sim p_1$ from the training data or jointly from a coupling as discussed in Section 2.2. Both x_t and $u_t(x_t|z)$ then follow directly from a closed-form conditional path given t and z which will be explained in the next paragraph. Finally $v_t(x_t)$ is simply a forward pass of the network being trained. Training therefore reduces to a plain regression problem.

There are many valid choices for the conditional path $p_t(x|z)$ and vector field $u_t(x|z)$ in Eq. (5). Throughout this paper we assume straight probability paths for all three coupling strategies discussed in Section 2.2. Each trajectory connects x_0 and x_1 along a straight line traversed at constant velocity

$$x_t = (1 - t)\, x_0 + t\, x_1, \qquad u_t(x_t|z) = x_1 - x_0 = \frac{x_1 - x_t}{1 - t}, \quad t \in [0, 1). \tag{6}$$

We work in the Euclidean space $\mathbb{R}^d$ where a straight line is the shortest connection between two points. Eq. (6) is therefore a reasonable default and is commonly referred to as the "optimal transport" (OT) path. The second equivalent form of u_t in Eq. (6) reappears directly in the velocity corrections used for guidance in Section 3. This choice is not forced by the framework itself. If the data instead lived on a curved manifold, a straight line in the ambient Euclidean space could leave it. Riemannian Flow Matching then replaces Eq. (6) with geodesics (the shortest paths on such spaces) (Chen and Lipman, 2024). Some applications instead call for a stochastic rather than a deterministic generative process. The conditional path can then be built from a Schrödinger bridge instead. This keeps an entropy-regularized transport structure closely related to the entropic coupling introduced in Section 2.2 while allowing genuine randomness along the trajectory (Tong et al., 2024).

2.2 Coupling

Equation (6) only fixes the shape of the path once the pair z is given. It says nothing about how x_0 and x_1 are paired up in the first place. That second choice is called the *coupling* π. A coupling π is any joint distribution over (x_0, x_1) with marginals p_0 and p_1 from which $z \sim \pi$ is drawn instead of the independent sampling $x_0 \sim p_0$ and $x_1 \sim p_1$ used above (Tong et al., 2024). Only the joint distribution π that produces the pair z changes between the three coupling strategies discussed below. The same linear path from Eq. (6) is used throughout. Figure 1 compares the resulting trajectories.

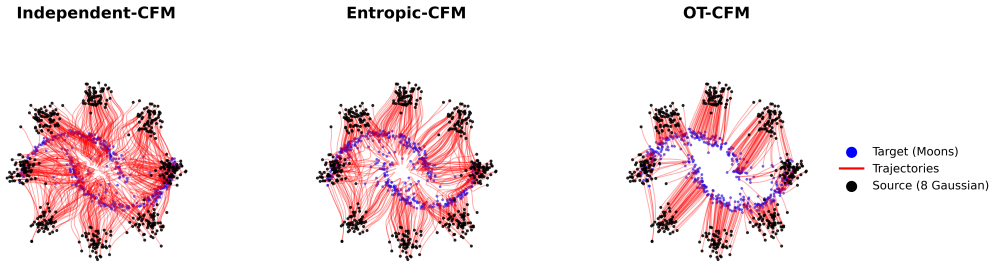


Figure 1: Comparison of three coupling strategies for flow matching: Independent-CFM, Entropic-CFM, and OT-CFM. Each panel shows sample trajectories (red) from the source distribution (8 Gaussians, black) to the target distribution (Moons, blue). OT-CFM produces shorter, straighter trajectories by minimizing transport cost, while Independent-CFM yields crossing paths.

2.2.1 Optimal Transport (OT) Coupling

The most natural choice follows directly from the motivation above. The OT coupling is the coupling π with marginal distributions p_0 and p_1 that minimizes the expected squared transport cost between x_0 and x_1

$$\pi^\star = \arg\min_{\pi} \; \mathbb{E}_{(x_0,x_1)\sim\pi} \left[\|x_1 - x_0\|^2 \right]. \tag{7}$$

We use the *minibatch OT* coupling of Tong et al. (2024): for each training batch of n source and n target points, Eq. (7) is solved exactly on just these $2n$ points with the network simplex algorithm, giving the cheapest one-to-one pairing within the batch. Sampling $z \sim \pi^\star$ then pairs each source point with the target point it can reach most cheaply, so trajectories stay short and rarely cross, as seen in the OT-CFM panel of Fig. 1. Such straight non-crossing paths are easier for v_t to approximate and require fewer ODE integration steps at inference (Tong et al., 2024).

2.2.2 Independent Coupling

The simplest and cheapest alternative avoids solving Eq. (7) altogether by taking the independent coupling $\pi(x_0, x_1) = p_0(x_0)\, p_1(x_1)$ (Tong et al., 2024). This pairs source and target points at random (Independent-CFM panel of Fig. 1). Because the pairing ignores distance, a source point can end up paired with a far-away unrelated target point. Trajectories can then cross one another and place conflicting demands on v_t at the same (t, x). Since v_t must assign a single velocity to every point (t, x), it cannot follow two crossing trajectories at once and instead ends up learning something close to their mean direction, distorting both. This makes the vector field harder to learn and typically requires more ODE integration steps at inference.

2.2.3 Entropic Coupling

Exact OT couplings are costly to recompute for every minibatch and can be unstable to estimate from a finite sample of an empirical distribution. The *entropic coupling* trades some of the straightness of Eq. (7) for robustness and speed by relaxing the transport problem with an added entropy term on π, weighted by a regularization strength $\lambda \in (0, \infty)$. This regularized problem is solved efficiently with the Sinkhorn algorithm rather than an exact assignment solver. As $\lambda \to 0$ the entropic coupling recovers the exact OT coupling, and as $\lambda \to \infty$ it approaches the independent coupling. For instance, we will use $\lambda \in \{0.1, 1.0, 10.0\}$ in our experiments. The intermediate trajectories of the Entropic-CFM panel of Fig. 1 reflect this. This construction extends naturally to unpaired settings such as single-cell genomics (Klein et al., 2024).

The coupling overall controls how straight and non-crossing the training trajectories are. Section 4.2 shows that this choice directly affects how well an already trained flow can be steered by inference-time guidance.

2.3 Class-conditional Flow Matching (CCFM)

The guidance methods introduced in Section 3 steer a flow matching model $v_t(x)$ at inference time, without the model ever seeing the desired subdistribution during training. As a training-time alternative, we consider Class-conditional Flow Matching (CCFM), which instead conditions the velocity field on a discrete label, analogous to conditional generation in diffusion models (Ho and Salimans, 2022) and to class-conditional flow-based generative models trained on labeled image data (Ma et al., 2024). Each target sample x_1 is assigned a label $y = c(x_1) \in \{1, \dots, K\}$ by a fixed partition c of the target distribution. Crucially, y requires no additional sampling step. It is a deterministic function of x_1, which is already part of the pair $z = (x_0, x_1)$ drawn for the ordinary CFM loss, so y is fixed as soon as z is, before x_t is drawn from the conditional path $p_t(\cdot|z)$. The conditional velocity field $v_t(x, y)$ is trained by minimizing

$$\mathcal{L}_{CCFM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, z=(x_0,x_1)\sim p(z),\, x_t\sim p_t(\cdot|z)} \left[\| v_t(x_t, c(x_1)) - u_t(x_t|z) \|^2 \right], \tag{8}$$

which is identical to Eq. (5) except that the network additionally receives the label $c(x_1)$ as input. The coupling and path construction of Section 2.2 are unaffected, but the label must be carried through the reshuffling induced by couplings so that $c(x_1)$ stays attached to its source-target pair z.

At inference, a sample from a desired subdistribution is generated by drawing y from the label distribution of that subdistribution and integrating Eq. (1) with $v_t(\cdot, y)$ in place of $v_t(\cdot)$, i.e., no test-time modification of the vector field is required. This makes CCFM substantially less general than the guidance methods of Section 3. It requires the label partition c to be fixed in advance and a separate model to be retrained whenever the target subdistribution changes, and it can only reach subdistributions expressible as a mixture of the fixed classes. In exchange, because the conditioning is learned rather than approximated at inference, we use CCFM in Section 4 as an oracle baseline against which the training-free guidance methods are compared.

3 Guidance

This section presents the guidance methods for flow matching introduced by Feng et al. (2025). Guidance modifies an already trained flow matching model at inference time so that it generates samples from a desired subdistribution of the target distribution p_1 without retraining the underlying model. Following Feng et al. (2025), we look for a guidance vector field $g_t(x_t)$ that is added to the trained velocity field, $v'_t(x_t) = v_t(x_t) + g_t(x_t)$, so that the modified field transports p_0 to the desired subdistribution instead of p_1. Which subdistribution is desired is expressed through a scalar energy function J, mapping $\mathbb{R}^d$ to $\mathbb{R}$, that assigns low values to preferred outcomes and high values to everything else, so that only the low-energy region is effectively generated. A concrete example of such an energy function, together with the subdistribution of p_1 it selects in our experiments, is given in Section 4. Low-energy regions are exponentially up-weighted while the rest of p_1 is left untouched, so J can encode anything from selecting a subset of the modes of p_1 to steering generation toward a desired property more generally. Formally, this amounts to sampling from a tilted target distribution

$$p'(x) = \frac{1}{Z}\, p_1(x)\, e^{-J(x)}, \qquad Z = \int p_1(x)\, e^{-J(x)}\, dx, \tag{9}$$

where Z is simply the normalizing constant that makes p' a valid probability distribution. Producing samples from p' instead of p_1 also means deciding how each new x_1 is paired with a source point x_0. Analogously to p' above, reweighting the joint coupling $\pi(x_0, x_1)$ used during training defines a new joint coupling for the guided process,

$$\pi'(x_0, x_1) = \frac{1}{Z}\, \pi(x_0, x_1)\, e^{-J(x_1)}. \tag{10}$$

Feng et al. (2025) compare the reverse coupling implied by π' to the one implied by the training coupling π through the reverse coupling ratio

$$\mathcal{P} = \frac{\pi'(x_0|x_1)}{\pi(x_0|x_1)}. \tag{11}$$

This ratio is exactly one for the independent coupling of Section 2.2.2, since there $\pi(x_0|x_1) = p_0(x_0)$ does not depend on x_1, and only approximately one for the OT and entropic couplings, where the pairing does depend on x_1. Accounting for $\mathcal{P}$, the exact guidance field takes the form

$$\begin{aligned} g_t(x_t) &= \int \left(\mathcal{P}\, \frac{e^{-J(x_1)}}{Z_t(x_t)} - 1 \right) u_t(x_t|z)\, p(z|x_t)\, dz \\ &= \mathbb{E}_{z\sim p(z|x_t)} \left[\left(\mathcal{P}\, \frac{e^{-J(x_1)}}{Z_t(x_t)} - 1 \right) u_t(x_t|z) \right], \\ Z_t(x_t) &= \int \mathcal{P}\, e^{-J(x_1)}\, p(z|x_t)\, dz \\ &= \mathbb{E}_{z\sim p(z|x_t)} \left[\mathcal{P}\, e^{-J(x_1)} \right], \end{aligned} \tag{12}$$

where the integral can be rewritten as an expectation over the posterior $p(z|x_t)$ of the latent variable $z = (x_0, x_1)$ given the current state x_t. Following Feng et al. (2025), we assume $\mathcal{P} = 1$ for the remainder of this section, which simplifies Eq. (12). This expression still cannot be evaluated directly, even under $\mathcal{P} = 1$. The posterior $p(z|x_t)$, the distribution over endpoints $z = (x_0, x_1)$ that are consistent with the current state x_t, has no closed form. The four methods below differ only in how they approximate this posterior or the resulting expectation, trading accuracy for computational cost, and we return to the effect of the $\mathcal{P} = 1$ assumption itself in Section 4.2.

3.1 Monte Carlo (MC) estimation

MC guidance replaces the expectation over the intractable posterior $p(z|x_t)$ with self-normalized importance sampling from the tractable training coupling $p(z)$. Drawing pairs $z^i = (x_0^i, x_1^i) \sim p(z)$ and reweighting by the known conditional density $p_t(x_t|z)$ gives the estimator

$$g_t^{MC}(x_t) = \mathbb{E}_{z\sim p(z)} \left[\left(\frac{e^{-J(x_1)}}{\hat{Z}_t} - 1 \right) u_t(x_t|z)\, \frac{p_t(x_t|z)}{p_t(x_t)} \right], \qquad p_t(x_t) = \mathbb{E}_{z\sim p(z)} \big[p_t(x_t|z) \big], \tag{13}$$

with $\hat{Z}_t$ estimated the same way from the same samples. As the number of samples grows, this estimator converges to the exact field of Eq. (**??**), so g^{MC} makes no assumption on the source distribution or on the coupling. Its price is variance. Once J is even moderately selective, most sampled x_1^i receive a near-zero weight $e^{-J(x_1^i)}$, so a large number of samples is needed for a stable estimate, and this requirement grows quickly with the dimensionality of the data.

3.2 Simulated-based Monte Carlo estimation

Rather than sampling from the training coupling and hoping enough samples land near the low-energy region of J, simulated (sim-)MC guidance approximates the posterior itself with a Gaussian $\tilde{p}(x_1|x_t) = \mathcal{N}(x_1;\, \hat{x}_1(x_t), \Sigma_t(x_t))$ fitted to match its first two moments, and draws samples directly from it,

$$g_t^{\text{sim-MC}}(x_t) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{e^{-J(x_1^i)}}{\tilde{Z}_t} - 1\right) u_t(x_t|z^i), \qquad x_1^i \sim \tilde{p}(x_1|x_t). \tag{14}$$

Because samples are now drawn already concentrated near the relevant region, far fewer of them are needed than for g^{MC}. This comes at the cost of a bias introduced by the Gaussian approximation, which can be poor when the true posterior $p(x_1|x_t)$ is multimodal.

3.3 Covariance Guidance (cov-A / cov-G)

A cheaper alternative common in the diffusion literature avoids sampling altogether. If the posterior $p(x_1|x_t)$ is assumed to be concentrated around its mean $\hat{x}_1(x_t) := \mathbb{E}_{z\sim p(z|x_t)}[x_1]$, which under the straight-line path of Eq. (6) is obtained cheaply from the current network prediction as $\hat{x}_1(x_t) = x_t + (1-t)\, v_t(x_t)$, a first-order Taylor expansion of $e^{-J(x_1)}$ around $\hat{x}_1$ turns the intractable expectation of Eq. (**??**) into a gradient of J evaluated at this single point estimate, preconditioned by the covariance $\Sigma_{1|t}$ of $p(x_1|x_t)$,

$$g_t^{cov}(x_t) \propto \Sigma_{1|t}\, \nabla_{\hat{x}_1} J(\hat{x}_1). \tag{15}$$

The two variants considered here differ only in how $\Sigma_{1|t}$ is obtained. **cov-A** replaces it with a hand-tuned decaying schedule $\lambda_t I$, so the guidance reduces to a rescaled gradient of J that requires no extra network evaluations. **cov-G** instead computes $\Sigma_{1|t}$ exactly under the additional assumption that the source distribution is Gaussian and independent of the target, the special case in which flow matching coincides with a diffusion model, using the Jacobian of v_t with respect to x_t. This recovers classical diffusion guidance methods such as DPS and LGD as a special case, but it requires backpropagating through the network at every sampling step.

3.4 Learned guidance (GM)

The methods above are training-free and are evaluated at sampling time on top of a fixed base model. Learned guidance instead amortizes the cost of guidance into a one-time training step. A network g_ϕ, together with an auxiliary network $Z_\phi \approx Z_t$, is fitted with a regression loss whose minimizer provably coincides with the exact field of Eq. (**??**),

$$\mathcal{L}_\phi = \mathbb{E}_{t,\,z,\,x_t}\left[\left\| g_\phi(x_t,t) - \left(\tfrac{e^{-J(x_1)}}{Z_\phi(x_t,t)} - 1\right) u_t(x_t|z)\right\|_2^2\right], \tag{16}$$

mirroring how the conditional flow matching loss of Eq. (5) replaces an intractable marginal objective with a tractable conditional one. Once trained, g_ϕ is a single cheap forward pass at inference time. The trade-off is that a new pair of networks must be trained for every energy function J, unlike the training-free methods above, which can be reused across arbitrarily many choices of J on the same base model.

3.5 Discussion

The four methods trade off bias, variance, and computational cost differently. g^{MC} makes the fewest assumptions and needs no additional training, but its variance grows quickly with the dimensionality of the data and the selectivity of J. Learned guidance removes this variance by shifting the cost to a one-time training step, at the price of retraining whenever J changes. The covariance methods are the cheapest at inference and need no sampling at all, but they rest on the assumption that $p(x_1|x_t)$ is

well summarized by a point estimate and a covariance, an assumption that can fail once J carves out a small or oddly shaped region of p_1. We compare all methods empirically in Section 4 and examine in Section 4.2 how their accuracy depends on the strength of the coupling used during training.

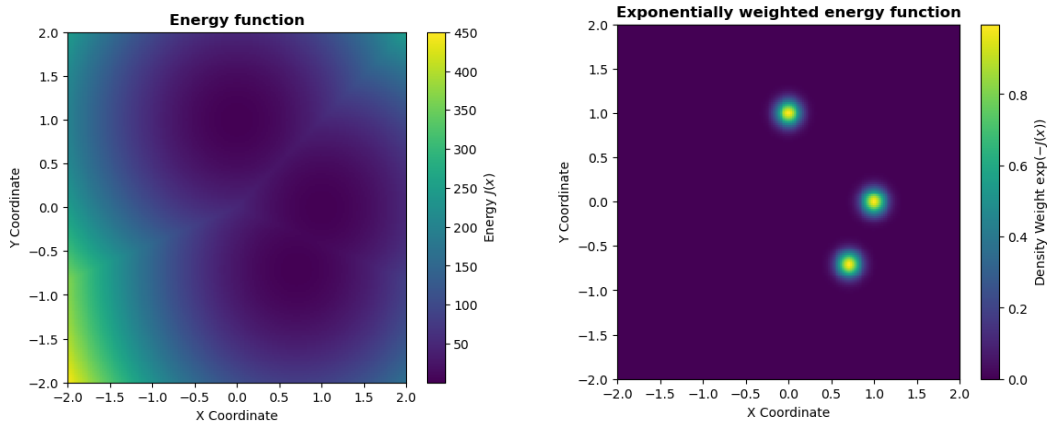
4 Results

4.1 Comparison of guidance methods

We define the energy function J to select three of the eight modes of the target distribution. For this, we define the energy function J as the sum of the squared distances to the three selected modes, scaled by a factor σ^2, which is the energy width.

$$J(x) = -\log \sum_{i \in 0,2,3} \exp\left(-\frac{|x - c_i|_2^2}{2\sigma^2}\right)$$

$$J(x) = \frac{|x - c_0|_2^2}{2\sigma^2} + \frac{|x - c_2|_2^2}{2\sigma^2} + \frac{|x - c_3|_2^2}{2\sigma^2}$$



(a) Energy function $J(x)$.

(b) Exponentially weighted energy $e^{-J(x)}$.

Figure 2: Left: the energy function $J(x)$ selecting three of the eight Gaussian modes (low values indicate preferred regions). Right: the corresponding exponentially weighted density $e^{-J(x)}$, proportional to the guided target $p'(x) \propto p_1(x)\, e^{-J(x)}$, with the three target modes clearly localized.

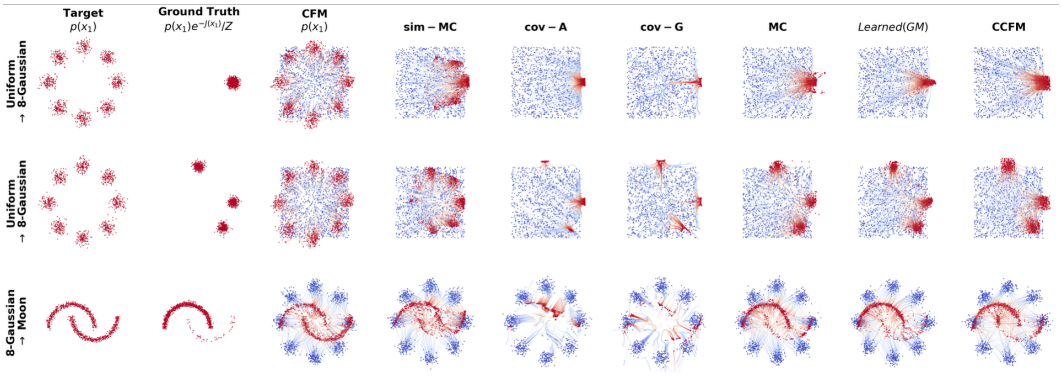


Figure 3: Qualitative comparison of guidance methods across three energy functions (rows). Columns show, from left to right: target distribution, ground truth $p'(x)$, and samples generated by each guidance method (CFM, sim-MC, cov-A, cov-G, MC, Learned (GM), CCFM). Red dots indicate samples in the guided target region; blue dots are the remaining samples.

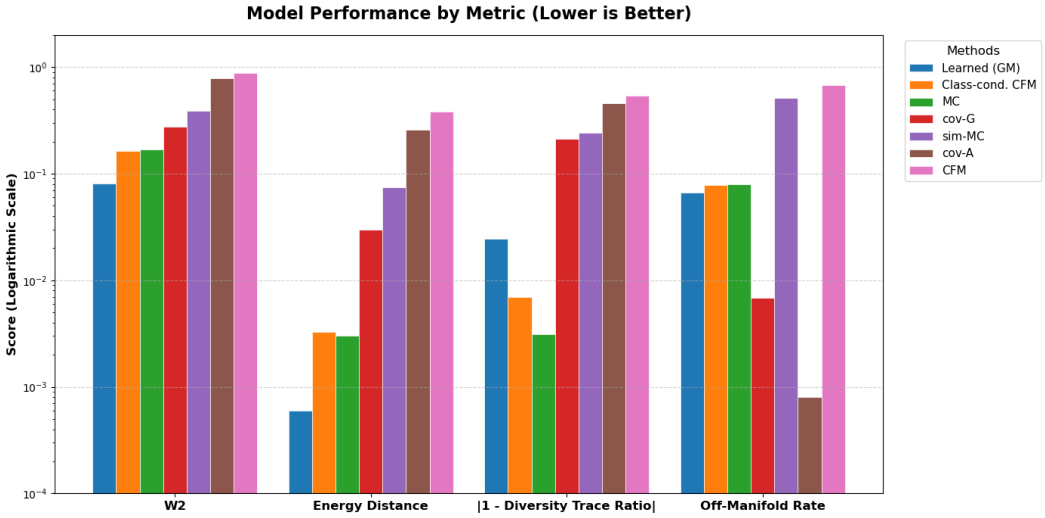
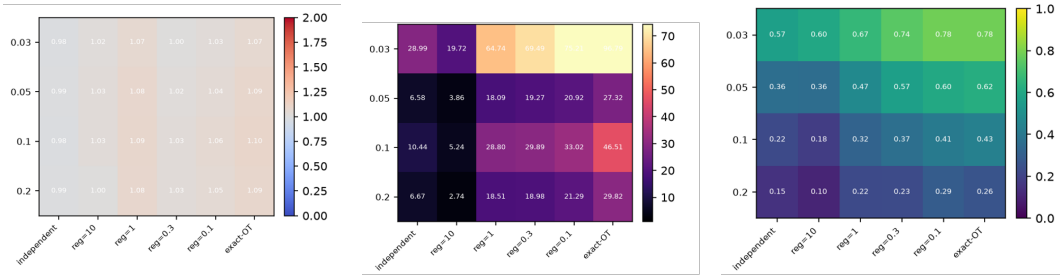


Figure 4: Quantitative comparison of all guidance methods on four metrics shown on a logarithmic scale. Lower is better for all metrics: W2 distance, energy distance, $|1 -$ diversity trace ratio$|$, and off-manifold rate. The learned (GM) model achieves the lowest energy distance; MC guidance offers a strong balance across all metrics without requiring additional training.

4.2 Limitations of guidance

There are several limitations of guidance methods. Since the MC method is the best inference-steering method, we focus on the limitations of the MC method in this section. In many applications, the OT-coupling is used to learn the velocity field



(a) Diversity trace ratio. (b) Energy distance. (c) Off-manifold rate.

Figure 5: Phase diagrams for MC guidance across coupling regularization strengths (columns, from independent coupling to exact-OT) and the stabilization parameter ϵ (rows, from 0.03 to 0.2). Stronger coupling (right columns) consistently increases the energy distance and off-manifold rate, indicating that the independence assumption of MC guidance breaks down under tight couplings.

4.3 Tuning hyperparameters

We want to discuss the effects of the hyperparameters on the guidance methods.

4.3.1 ODE solver

We increased the number of steps in the ODE solver from 100 to 200. This increased the performance of the guidance methods. Moreover, one could use a different ODE solver, e.g. Runge-Kutta 4th order, which might improve the generated samples further with the downside of increased computational cost. An early stopping criterion could be used to mitingate the problem of numerical instabilities at $t = 1$ and mode collapse in the MC guidance method.

4.3.2 MC Guidance

The parameter `mc_ep` (ϵ) appears in the denominator $(1 - t + \epsilon)$ of both the MC importance weights and the velocity correction $u = (x_1 - x_t)/(1 - t + \epsilon)$, acting as a numerical stabilizer as $t \to 1$. Without sufficient ϵ, back-projected source points fall outside the prior support, causing all importance weights to clamp to a uniform distribution and injecting large random forces that scatter particles off the manifold. We tried $\epsilon \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$, with results shown in Table 1. Hence, we increased `mc_ep` from 0.001 to 0.05, which reduced the numerical instability and the outliers of the generated distribution significantly.

Table 1: Effect of the `mc_ep` parameter ϵ on sample quality.

ϵ	W_2 Distance ↓	Off-Manifold Rate ↓
0.001 (default)	0.4244	0.266
0.010	0.3053	0.042
0.050 (tuned)	**0.1091**	**0.002**
0.100	0.2032	0.004

4.3.3 Cov Guidance

5 Conclusion

Conclusion

$p'(x) = \frac{1}{Z} p_1(x) e^{-J(x)}$

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A Appendix

A.1 Result data in tabular form

Table 2: Comparison of methods across various metrics. The best method for each metric is **bolded**, and the second best is underlined.

Method	W2 ↓	Energy Dist ↓	Div Trace Ratio	Off-Manifold Rate ↓
Learned (GM)	**0.0802**	**0.0006**	0.9757	0.0659
Class-cond. CFM	0.1632	0.0033	1.0069	0.0786
MC	0.1698	0.0030	**0.9969**	0.0801
cov-G	0.2772	0.0296	1.2113	0.0068
sim-MC	0.3908	0.0739	0.7570	0.5098
cov-A	0.7905	0.2578	0.5410	**0.0008**
CFM	0.8852	0.3804	1.5355	0.6782